# Marvin Minsky

**Marvin Lee Minsky** (August 9, 1927 – January 24, 2016) was an American cognitive and computer scientist concerned largely with research in artificial intelligence (AI). He co-founded the Massachusetts Institute of Technology's AI laboratory and wrote extensively about AI and philosophy.[12][13][14][15]

Minsky received many accolades and honors, including the 1969 ACM Turing Award. He is known as the "father of AI".[16]

| Marvin Minsky | |
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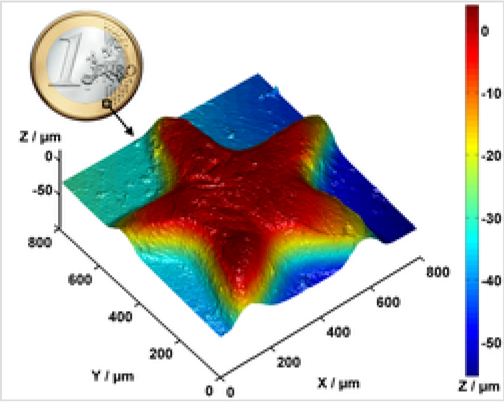
|  | |
| Minsky in 2008 | |
| **Born** | Marvin Lee Minsky<br>August 9, 1927<br>New York City, U.S. |
| **Died** | January 24, 2016 (aged 88)<br>Boston, Massachusetts, U.S. |
| **Education** | Harvard University (BA)<br>Princeton University (MA, PhD) |
| **Known for** | Artificial intelligence[5]<br>Confocal microscope[6]<br>Useless machine[7]<br>Triadex Muse[8]<br>*Perceptrons*[9]<br>*The Society of Mind*[10]<br>*The Emotion Machine*[11]<br>Frames<br>SNARC<br>Dartmouth workshop |
| **Spouse** | Gloria Rudisch (m. 1952) |
| **Children** | 3 |

## Early life and education

Marvin Lee Minsky was born in New York City, to Henry, an eye surgeon, and Fannie (Reiser), a Zionist activist.[15][17][18] His family was Jewish. He attended the Ethical Culture Fieldston School and the Bronx High School of Science. He later attended Phillips Academy in Andover, Massachusetts. He then served in the US Navy from 1944 to 1945. He received a B.A. in mathematics from Harvard University in 1950 and a Ph.D. in mathematics from Princeton University in 1954. His doctoral dissertation was titled "Theory of neural-analog reinforcement systems and its application to the brain-model problem."[19][20][21] He was a Junior Fellow of the Harvard Society of Fellows from 1954 to 1957.[22][23]

Minsky was on the MIT faculty from 1958 to his death. He joined the staff at MIT Lincoln Laboratory in 1958; a year later, he and John McCarthy initiated what was, as of 2003, named the MIT Computer Science and Artificial Intelligence Laboratory.[24][25] He was the Toshiba Professor of Media Arts and Sciences as well as professor of electrical engineering and computer science at MIT.

## Contributions in computer science




3D profile of a coin (partial) measured with a modern confocal white light microscope

Minsky's inventions include the first head-mounted graphical display (1963)[26] and the confocal microscope[6][note 1] (1957, a predecessor to today's widely used confocal laser scanning microscope). With Seymour Papert, he developed the first Logo "turtle". In 1951, Minsky built the first randomly wired neural network learning machine, SNARC. In 1962, he worked on small universal Turing machines and published his well-known 7-state, 4-symbol machine.[27]

Minsky's book *Perceptrons* (written with Papert) attacked the work of Frank Rosenblatt, and became the foundational work in the analysis of artificial neural networks. The book is the center of a controversy in the history of AI, as some claim it greatly discouraged research on neural networks in the 1970s and contributed to the so-called "AI winter".[28] Minsky also founded several other AI models. His paper "A framework for representing knowledge"[29] created a new paradigm in knowledge representation. *Perceptrons* is now more a historical than practical book, but the theory of frames is in wide use.[30] Minsky also wrote of the possibility that extraterrestrial life may think like humans, thus permitting communication.[31]

In the early 1970s, at the MIT Artificial Intelligence Lab, Minsky and Papert started developing what came to be known as the Society of Mind theory. The theory

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| **Awards** | Turing Award (1969)<br>Japan Prize (1990)<br>AAAI Fellow (1990)[1]<br>IJCAI Award for Research Excellence (1991)<br>Benjamin Franklin Medal (2001)<br>BBVA Foundation Frontiers of Knowledge Award (2013) |
| | **Scientific career** |
| **Fields** | Cognitive science<br>Computer science<br>Artificial intelligence<br>Philosophy of mind |
| **Institutions** | Massachusetts Institute of Technology |
| **Thesis** | *Theory of Neural-Analog Reinforcement Systems and Its Application to the Brain Model Problem (https://www.proquest.com/docview/301998727)* (1954) |
| **Doctoral advisor** | Albert W. Tucker[2][3] |
| **Doctoral students** | James Robert Slagle<br>Manuel Blum<br>Daniel Bobrow<br>Ivan Sutherland<br>Bertram Raphael<br>William A. Martin<br>Joel Moses<br>Warren Teitelman<br>Adolfo Guzmán Arenas<br>Patrick Winston<br>Eugene Charniak<br>Gerald Jay Sussman<br>Scott Fahlman<br>Benjamin Kuipers<br>Luc Steels<br>Danny Hillis<br>K. Eric Drexler<br>Berthold K.P. Horn |

attempts to explain how what we call intelligence could be a product of the interaction of non-intelligent parts. Minsky says that the biggest source of ideas for the theory came from his work in trying to create a machine that uses a robotic arm, a videocamera, and a computer to build with children's blocks. In 1986, he published *The Society of Mind*, a comprehensive book on the theory which, unlike most of his previously published work, was written for the general public.

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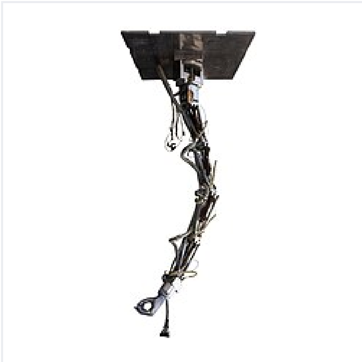
|---|---|
| | Carl Hewitt<br>David Levitt[4] |
| **Website** | web.media.mit.edu/~minsky (http://web.media.mit.edu/~minsky) |

**The MA-3 Robotic Manipulator Arm, on display at MIT Museum**



General view

The Belgrade Hand

In 2006, Minsky published *The Emotion Machine*, a book that critiques many popular theories of how human minds work and suggests alternative theories, often replacing simple ideas with more complex ones.[32]

Minsky also invented a "gravity machine" that will ring a bell if the gravitational constant changes, a theoretical possibility that is not expected to occur in the foreseeable future.[7]

# Role in popular culture

Minsky was an adviser[33] on Stanley Kubrick's movie *2001: A Space Odyssey*; one of the movie's characters, Victor Kaminski, was named in Minsky's honor.[34] Minsky is mentioned explicitly in Arthur C. Clarke's novel of the same name, where he is portrayed as achieving a crucial breakthrough in artificial intelligence in the then-future 1980s, paving the way for HAL 9000 in the early 21st century:

> In the 1980s, Minsky and Good had shown how artificial neural networks could be generated automatically—self replicated—in accordance with any arbitrary learning program. Artificial brains could be grown by a process strikingly analogous to the development of a human brain. In any given case, the precise details would never be known, and even if they were, they would be millions of times too complex for human understanding.[35]

In "The Law of Non-Contradiction", episode 3 of the television anthology series Fargo (Season 3), at least two allusions to Minsky are made. The first is through the depiction of a "useless machine": a device Minsky invented as a philosophical joke. Claude Shannon, Minsky's mentor at Bell Labs, built the first working prototype of this machine.[36] The second is through the depiction of an animation of a robot called "minsky"—a character in the sci-fi novel *The Planet Wyh*.

# Personal life

In 1952, Minsky married pediatrician Gloria Rudisch; together they had three children.[37] Minsky was a talented improvisational pianist[38] who published musings on the relations between music and psychology.

The Minskytron or "Three Position Display" running on the Computer History Museum's PDP-1, 2007

## Opinions

Minsky was an atheist.[39] He was a signatory to the Scientists' Open Letter on Cryonics.[40]

He was a critic of the Loebner Prize for conversational robots,[41] and argued that a fundamental difference between humans and machines is that while humans are machines, they are machines in which intelligence emerges from the interplay of the many unintelligent but semi-autonomous agents the brain comprises.[42] He argued that "somewhere down the line, some computers will become more intelligent than most people", but that it was very hard to predict how fast progress would be.[43] He cautioned that an artificial superintelligence designed to solve an innocuous mathematical problem might decide to assume control of Earth's resources to build supercomputers to help achieve its goal,[44] but believed that such scenarios are "hard to take seriously" because he felt confident that AI would be well tested before being deployed.[45]

## Association with Jeffrey Epstein

Minsky received a $100,000 research grant from Jeffrey Epstein in 2002, four years before Epstein's first arrest for sex offenses; it was the first from Epstein to MIT. Minsky received no further research grants from him.[46][47]

Minsky organized two academic symposia on Epstein's private island Little Saint James, one in 2002 and another in 2011, after Epstein was a registered sex offender.[48] Virginia Giuffre testified in a 2015 deposition in her defamation lawsuit against Epstein's associate Ghislaine Maxwell that Maxwell "directed" her to have sex with Minsky, among others. Giuffre writes in her memoir, "Epstein sent me to a cabana on the beach and told me to service the man inside. I will never forget Minsky’s bald head, and the way his face seemed to have shriveled like one of those folk-art dolls whose heads are dried-up apples. Throughout my time having sex with Minsky, I could hear the waves lapping outside the little room. I tried to focus only on that sound."[49] There has been no lawsuit against Minsky's estate.[50] Minsky's widow, Gloria Rudisch, says that he could not have had sex with any of the women at Epstein's residences, as they were always together during all the visits to Epstein's residences.[51] But in her posthumously published memoir, Giuffre writes that she was forced to have sex with Minsky.[52]

### Death

Minsky died of a cerebral hemorrhage in January 2016, at age 88.[53] Minsky was a member of Alcor Life Extension Foundation's Scientific Advisory Board.[54] Alcor will neither confirm nor deny whether Minsky was cryonically preserved.[55]

## Selected bibliography

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- 1969 – *Perceptrons: An Introduction to Computational Geometry*, MIT Press
- 1986 – *The Society of Mind*
- 2006 – *The Emotion Machine: Commonsense Thinking, Artificial Intelligence, and the Future of the Human Mind*

## Awards and affiliations

Minsky won the Turing Award (the greatest distinction in computer science)[42] in 1969, the Golden Plate Award of the American Academy of Achievement in 1982,[56] the Japan Prize in 1990,[57] the IJCAI Award for Research Excellence for 1991, and the Benjamin Franklin Medal from the Franklin Institute for 2001.[58] In 2006, he was inducted as a Fellow of the Computer History Museum "for co-founding the field of artificial intelligence, creating early neural networks and robots, and developing theories of human and machine cognition."[59] In 2011, Minsky was inducted into IEEE Intelligent Systems' AI Hall of Fame for the "significant contributions to the field of AI and intelligent systems".[60] In 2014, Minsky won the Dan David Prize for "Artificial Intelligence, the Digital Mind".[61] He was also awarded with the 2013 BBVA Foundation Frontiers of Knowledge Award in the Information and Communication Technologies category.[62]

Minsky was affiliated with the following organizations:

- United States National Academy of Engineering[26]
- United States National Academy of Sciences[26]
- Extropy Institute's Council of Advisors[63]
- Alcor Life Extension Foundation's Scientific Advisory Board[54]
- kynamatrix Research Network's Board of Directors[64]

## Media appearances

- *Machine Dreams* (1988)
- *Future Fantastic* (1996)

## See also

- List of pioneers in computer science

# Notes

1. The patent for Minsky's Microscopy Apparatus was applied for in 1957, and subsequently granted US Patent Number 3,013,467 in 1961. According to his published biography on the MIT Media Lab webpage, "In 1956, when a Junior Fellow at Harvard, Minsky invented and built the first Confocal Scanning Microscope, an optical instrument with unprecedented resolution and image quality".

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# External links

- Scientist on the Set: An Interview with Marvin Minsky (https://web.archive.org/web/20120716182537/http://mitpress.mit.edu/e-books/Hal/chap2/two1.html)
- Consciousness Is A Big Suitcase: A talk with Marvin Minsky (http://www.edge.org/3rd_culture/minsky/)
- Video of Minsky speaking at the International Conference on Complex Systems, hosted by the New England Complex Systems Institute (NECSI) (https://web.archive.org/web/20120320001744/http://necsi.edu/events/iccs/video/iccs2002wednesday/5-minskyclip.html)
- "The Emotion Universe": Video with Marvin Minsky (https://web.archive.org/web/20060829135040/http://www.edge.org/video/dsl/EF02_minsky.html)
- Marvin Minsky's thoughts on the Fermi Paradox at the Transvisions 2007 conference (http://sentientdevelopments.blogspot.com/2007/07/when-dvorsky-met-minsky.html)
- "Health, population and the human mind" (https://www.ted.com/talks/marvin_minsky_health_and_the_human_mind) : Marvin Minsky talk at the TED conference
- "The Society of Mind" (http://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-868j-the-society-of-mind-fall-2011/) on MIT OpenCourseWare
- Marvin Minsky (http://www.webofstories.com/people/marvin.minsky/1) tells his life story at Web of Stories (video)
- Marvin Minsky Playlist (http://web.mit.edu/echemi/www/031126.html) Archived (https://web.archive.org/web/20160210110026/http://web.mit.edu/echemi/www/031126.html) February 10, 2016, at the Wayback Machine Appearance on WMBR's *Dinnertime Sampler (http://web.mit.edu/echemi/www/index.html) Archived (https://web.archive.org/web/20110504000207/http://web.mit.edu/echemi/www/index.html) May 4, 2011, at the Wayback Machine* radio show November 26, 2003
- Oral history interview with Marvin Minsky (http://purl.umn.edu/107503) at Charles Babbage Institute, University of Minnesota, Minneapolis. Minsky describes artificial intelligence (AI) research at the Massachusetts Institute of Technology (MIT). Topics include: the work of John McCarthy; changes in the MIT research laboratories with the advent of Project MAC; research in the areas of expert systems, graphics, word processing, and time-sharing; variations in the Advanced Research Projects Agency (ARPA) attitude toward AI.

- Oral history interview with Terry Winograd (http://purl.umn.edu/107717) at Charles Babbage Institute, University of Minnesota, Minneapolis. Winograd describes his work in computer science, linguistics, and artificial intelligence at the Massachusetts Institute of Technology (MIT), discussing the work of Marvin Minsky and others.
- Appearances (https://www.c-span.org/person/?1008481) on C-SPAN

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